

Question	Marking Guidance	Mark	Comments
03.1	Flagellum/flagella;	1 (AO1)	

Question	Marking Guidance	Mark	Comments
03.2	1. 3D with SEM, but 2D with TEM OR (Only) surface visible with SEM, but internal structures visible with TEM; 2. (Because) electrons deflected/bounce off (using SEM) OR Electrons transmitted/pass through (using TEM);	2 (2 x AO1)	2. Accept Resolution is lower (with SEM)

Question	Marking Guidance	Mark	Comments
03.3	Shorter <u>wavelength</u> between electrons OR Longer <u>wavelength</u> in light (rays);	1 (AO1)	Accept 'of electrons' OR 'of electron beam' for 'between electrons' Reject Electron microscope has a shorter wavelength Reject Light microscope has a longer wavelength

Question	Marking Guidance	Mark	Comments
03.4	Mark in pairs: 1 and 2 OR 3 and 4 1. Measure (length of structure) and divide by magnification; 2. Correct conversion from measured length to μm, either $\times 10\,000$ from cm OR $\times 1000$ from mm; 3. Measure (length of structure) and divide by (image) length of scale bar; 4. Multiply by actual length of scale bar;	2 (2 x AO1)	1. Accept correct equation making reference to a measurement

Question	Marking Guidance	Mark	Comments
03.5	1. Circular DNA (molecule in cytoplasm); 2. <u>Murein</u> cell wall OR <u>Peptidoglycan</u> cell wall OR <u>Glycoprotein</u> cell wall; 3. Small(er)/70S ribosomes (in cytoplasm);	2 max (2 x AO1)	Accept mesosome

Question	Marking Guidance	Mark	Comments
03.6	1. Capsid; 2. Reverse transcriptase; 3. RNA genome; 4. Envelope;	2 max (2 x AO1)	1. Accept capsomere 1. Ignore protein coat 3. Accept 'genetic material' OR 'genes' for 'genome'